HD-EMG Discussion

10-2025

Context

Stroke: rapidly developed clinical signs of focal (or global) disturbance of cerebral function, lasting more than 24 hours or leading to death, with no apparent cause other than of vascular origin.

In 2022, stroke is the 1st cause of acquired disability in adults.

Over 10 million people suffering from it annually.

(According to World Health Organization and World Stroke Organization)

Context

About 50% of stroke survivors have impaired upper limb and hand function in the chronic phase, including manual dexterity impairment. [1, 2]

It has a strong impact on activities of daily living and on independence. [3, 4]

AI generated

Dexterity: the ability to perform accurate and coordinated hand and finger movements and relatively independent finger movements to grasp and manipulate objects. [5]

[1]: Kwakkel, Gert, et al. "Probability of regaining dexterity in the flaccid upper limb: impact of severity of paresis and time since onset in acute stroke." Stroke 34.9 (2003): 2181-2186.
[2]: Parker, V. M., D. T. Wade, and R. Langton Hewer. "Loss of arm function after stroke: measurement, frequency, and recovery." International rehabilitation medicine 8.2 (1986): 69-73.
[3]: Rule, Katherine, et al. "Purdue manual dexterity testing: A cohort study of community-dwelling elderly." Journal of Hand Therapy 34.1 (2021): 116-120.
[4]: Carment, Loic, et al. "Manual dexterity and aging: a pilot study disentangling sensorimotor from cognitive decline." Frontiers in neurology 9 (2018): 910.
[5]: Térémetz, Maxime, et al. "A novel method for the quantification of key components of manual dexterity after stroke." Journal of neuroengineering and rehabilitation 12 (2015): 1-16.

Dexterity recovery in stroke

The mechanisms of dexterity recovery after stroke require further investigation, as dexterity recovery remains an underexplored area in stroke rehabilitation research [1].

Dexterity recovery is particularly challenging for clinicians, as current approaches primarily focus on proximal muscle training, rather than isolated finger movements and fine motor control [2].

In stroke rehabilitation, tools that provide detailed feedback on individual finger movements and forces are limited [3, 4].

[1]: Caplan, B., et al. (2021). "Understanding fine motor control: Mechanisms and applications in rehabilitation." Neuroscience Research Journal, 45(2), 133-145.
[2]: Frere, J., & Hug, F. (2012). "Between-subject variability of muscle synergies during a complex motor skill." Frontiers in Human Neuroscience, 6, 8.
[3]. Gazzoni, M., et al. (2018). "Real-time EMG biofeedback for training finger independence in rehabilitation." Journal of NeuroEngineering and Rehabilitation, 15(1), 95.
[4]. Williams, T., et al. (2020). "Biofeedback interventions for hand dexterity post-stroke." Journal of Rehabilitation Research and Development, 57(3), 215-225.

Is the HD-EMG an efficient tool for stroke rehabilitation?

HD-EMG provides high spatial resolution, allowing detailed recording of muscle activation patterns [5].

It enables detection of muscle synergies and motor unit decomposition, making it a promising tool for stroke rehabilitation [6, 7].

Using HD-EMG for classifying hand and finger movements could offer significant benefits, especially for capturing complex coordination [8].

Existing studies primarily focus on classifying hand postures or multi-finger movements; however, research on isolated finger movements and individual finger forces is limited [9, 11, 12, 13].

[5]: Farina, D., et al. (2010). "Decoding the neural drive to muscles from the surface electromyogram." Clinical Neurophysiology, 121(10), 1616-1623.
[6]: Merletti, R., & Farina, D. (2009). "Surface electromyography: Physiology, engineering, and applications." IEEE Engineering in Medicine and Biology Magazine, 28(2), 27-37.
[7] Gazzoni, M., & Farina, D. (2011). "High-density EMG in stroke rehabilitation." Neurorehabilitation and Neural Repair, 25(6), 564-571.
[8]: Castellini, C., & van der Smagt, P. (2013). "Surface EMG in advanced hand prosthetics." Biological Cybernetics, 107(6), 419-427.
[9]: Cesqui, B., et al. (2011). "Classification of hand movements using high-density electromyography." Journal of NeuroEngineering and Rehabilitation, 8, 60
[11]: A database of high-density surface electromyogram signals comprising 65 isometric hand gestures, Nebojša Malešević et al. Nature, 2021
[12]: Transformer-based hand gesture recognition from instantaneous to fused neural decomposition of high-density EMG signals, Mansooreh Montazerin et al. Nature, 2023
[13]: Thesis : Deep Learning Methods for Hand Gesture Recognition via High-Density Surface Electromyogram (HD-sEMG) Signals. By Mansooreh Sadat Montazerin, 09/2023.

Can the HD EMG signals give easy feedback to interpret?

According to this review of Wei Li[14]: “The conversion of sEMG signals into effective control signals often requires a lot of computational power and complex process.”

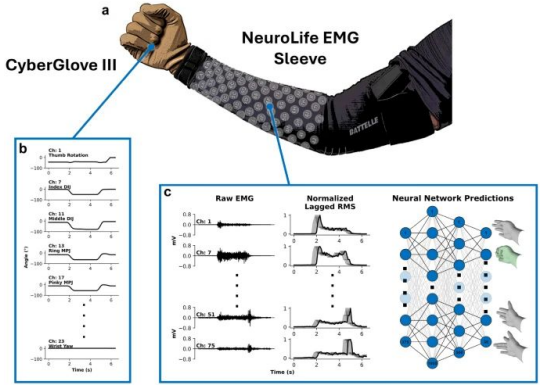
According to Tacca .N: “Stroke subjects, on average, had higher co-contraction and reduced muscle coupling, differences in coordination, overactivation of antagonist muscles and compensatory strategies.”

“The significant improvement of hardware equipment and the continuous emergence of large data sets of sEMG have also boosted the DL research in sEMG signal processing. DL can effectively improve the accuracy of sEMG pattern recognition and reduce the influence of interference factors.”

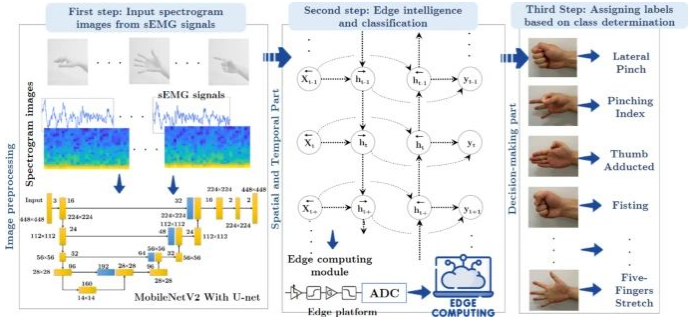
[14]: Li, W., Shi, P., & Yu, H. (2021). Gesture recognition using surface electromyography and deep learning for prostheses hand: state-of-the-art, challenges, and future. Frontiers in neuroscience, 15, 621885.
[15]: Tacca, N., Baumgart, I., Schlink, B. R., Kamath, A., Dunlap, C., Darrow, M. J., ... & Meyers, E. C. (2024). Identifying alterations in hand movement coordination from chronic stroke survivors using a wearable high-density EMG sleeve. Journal of Neural Engineering, 21(4), 046040.

How can we develop a learning-based model that delivers calibrated, real-time feedback?

State of the art



Wearable high-density EMG sleeve for complex hand gesture classification and continuous joint angle estimation [17]

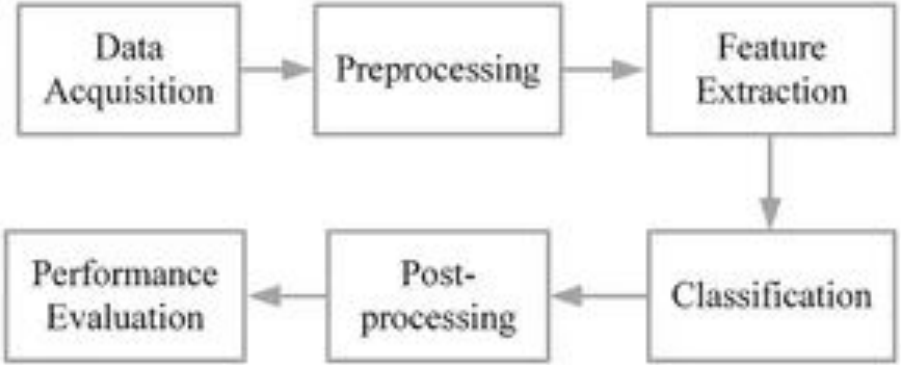


Hand gestures classification of sEMG signals based on BiLSTM-metaheuristic optimization and hybrid U-Net-MobileNetV2 encoder architecture [16]

[16]: Khavari, S. F., Rezaee, K., Ansari, M., Zare, F., & Roknabadi, M. H. A. (2024). Hand gestures classification of semg signals based on bilstm-metaheuristic optimization and hybrid u-net-mobilenetv2 encoder architecture.

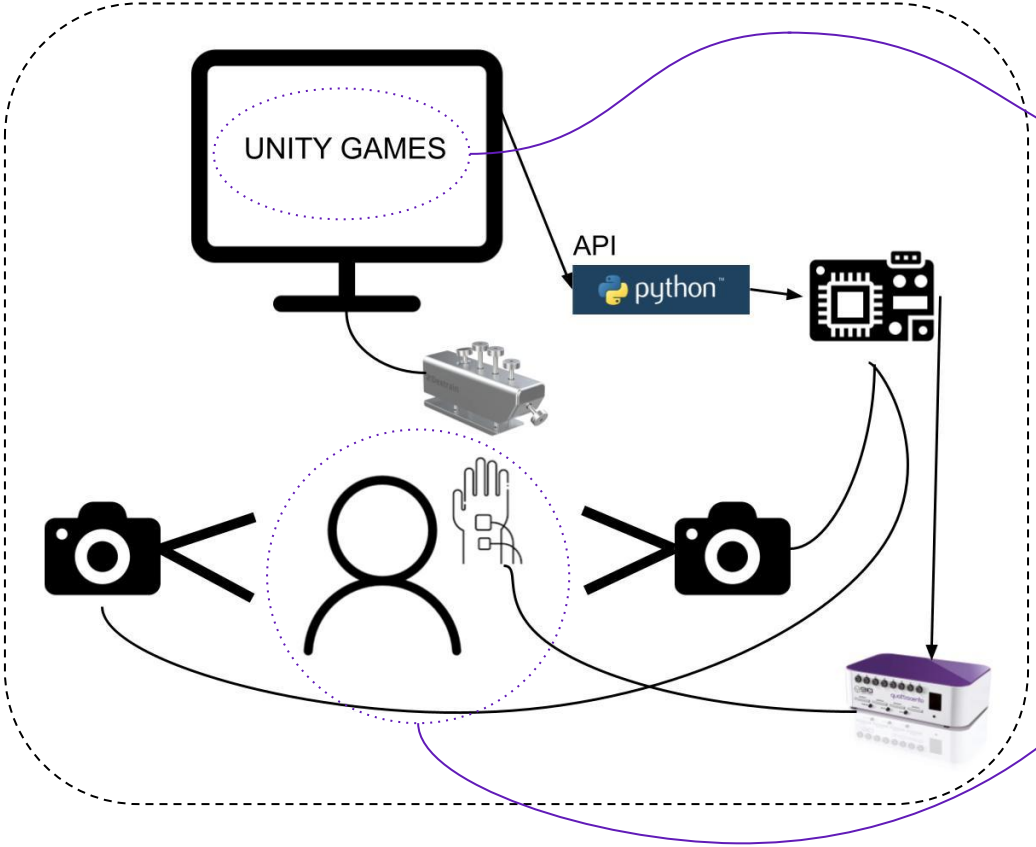
[17]: Tacca, N., Dunlap, C., Donegan, S. P., Hardin, J. O., Meyers, E., Darrow, M. J., ... & Friedenberg, D. A. (2024). Wearable high-density EMG sleeve for complex hand gesture classification and continuous joint angle estimation. Scientific Reports, 14(1), 18564.

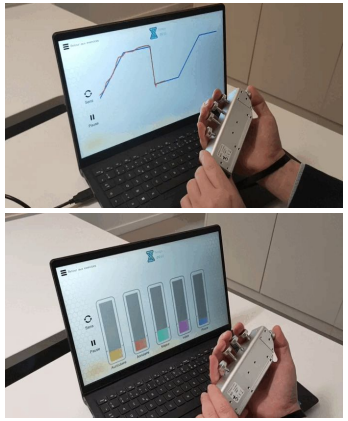
State of the art

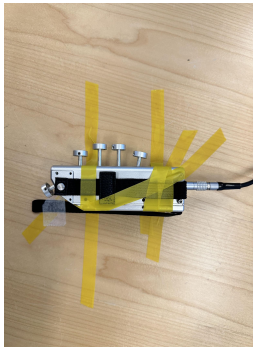


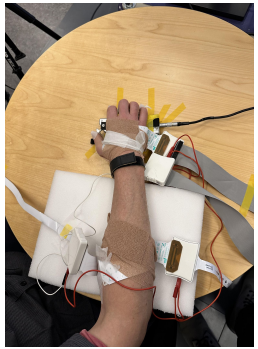
[14]: Li, W., Shi, P., & Yu, H. (2021). Gesture recognition using surface electromyography and deep learning for prostheses hand: state-of-the-art, challenges, and future. Frontiers in neuroscience, 15, 621885.

Method: data Acquisition





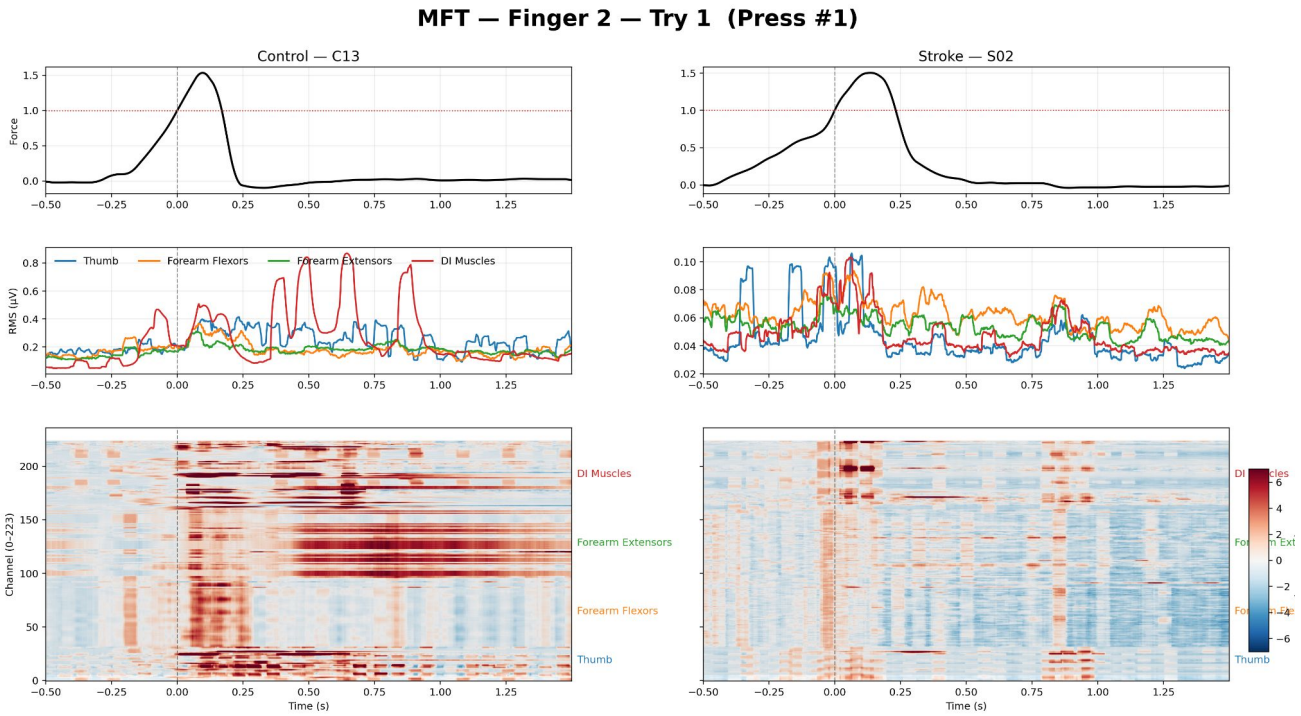




Results – Dataset

	Dataset
Tasks	Precision finger tasks (strength and movement)
Force range	0 to 9 N (<0.01 N accurate)
Subjects	17 controls + 4 stroke
Number of electrodes	224 electrodes

Raw data: what happens during one tap?



Tendance (no significative: not enough Stroke patients):

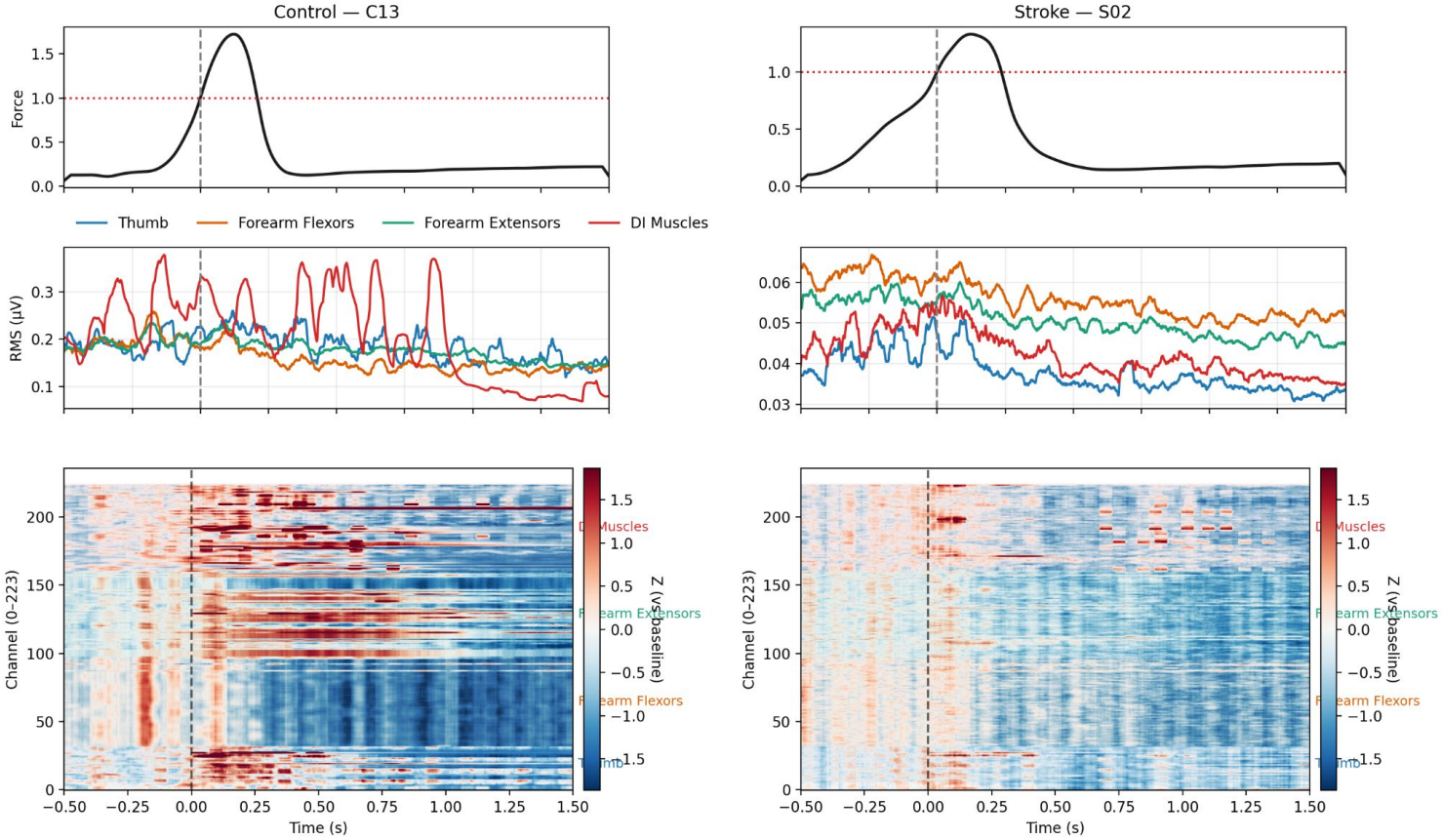
stroke cocontraction >
stroke selectivity channels <

Correlate with the lit.

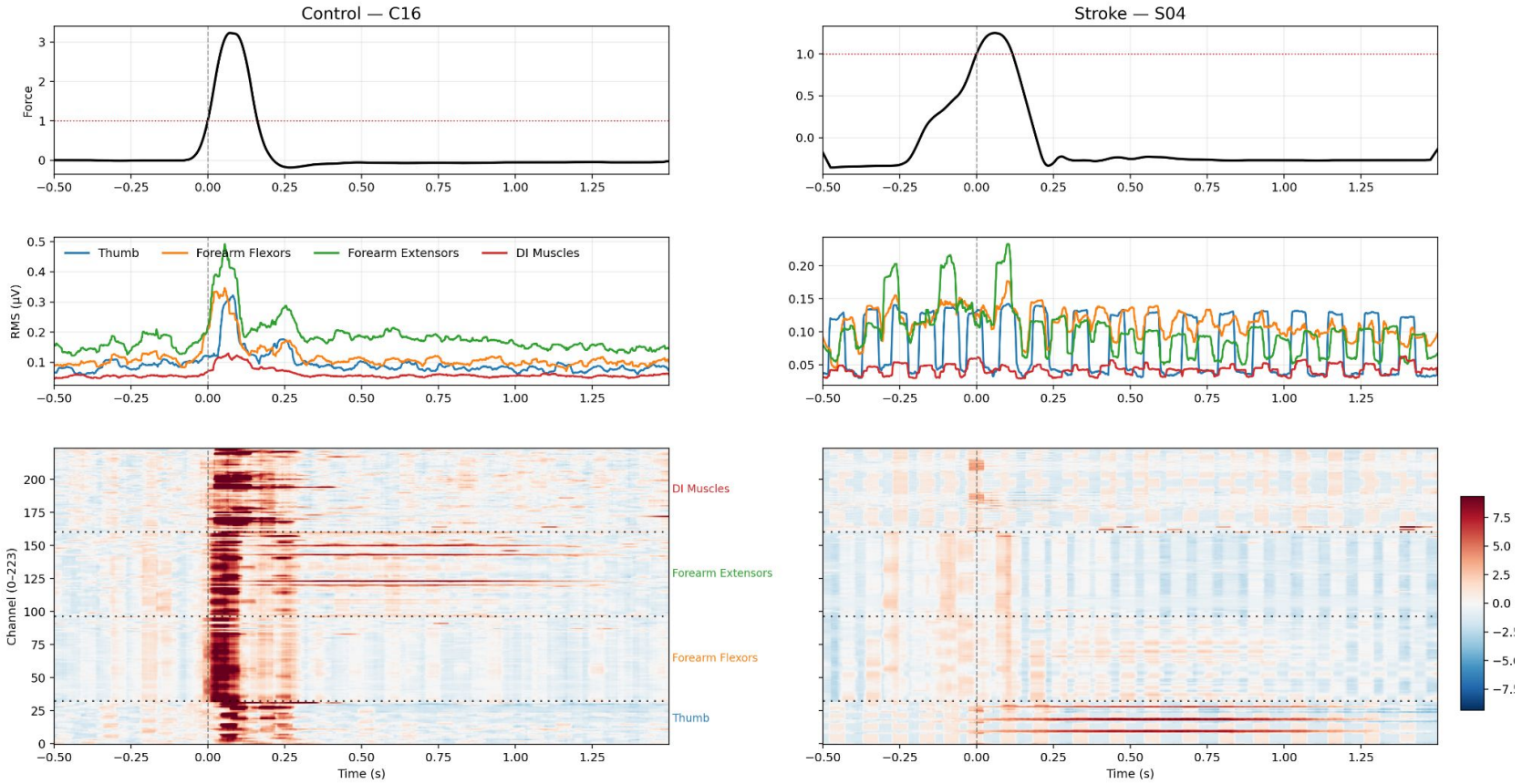
Signal hard to read: big variability inter and intra subjects. (also correlates with the lit)

Left: Control (C13) & right: Stroke (S02). Traces are time-locked to press onset.
Top: Reference force , red dotted line = detection threshold.
Middle: Grouped HD-EMG RMS (50 ms) for Thumb, Forearm Flexors, Forearm Extensors, DI (colors match labels).
Bottom: Per-channel RMS - Z-score heatmaps.

MFT — Finger 2 — Try 1 (Avg: Control 7 / Stroke 12 presses)



MFT — Finger 2 — Try 1 (Press #1)



Pre processing used in HD-sEMG DL pipelines

Cleaning:

- Notch 50/60 Hz, band-pass 20–400 Hz [14]
- Detect/repair bad channels: (abnormal amplitude / dead electrodes) and replace them by nearby neighbors to stabilize the deep-learning input on dense grids [20, 21, 22]

Remove rest:

- Drop windows where ref < 0.15 N; add ±150 ms padding around rest (not losing initiation phases). [14]

Windowing & RMS features:

- Sliding windows 100–200 ms, stride 25–50 ms; per-channel RMS.. [14, 17]

[20] Chen, W. et al., 2024 — Sensors: HD-sEMG bad-channel detection via absolute max value (threshold at channel-wise median) and neighbor replacement; 20–350 Hz band-pass + 50 Hz notch.
[21] Farago, F., & Chan, A., 2023 — Sensors: Local detection + 2D interpolation reconstruction using robust criteria (line-noise power, variance outliers) while preserving matrix size.
[22] Liu, X. et al., 2024 — (PMC) Hand Gesture Recognition with HD-sEMG: practical pipelines replace defective channels with the median/mean of surrounding electrodes to improve stability.

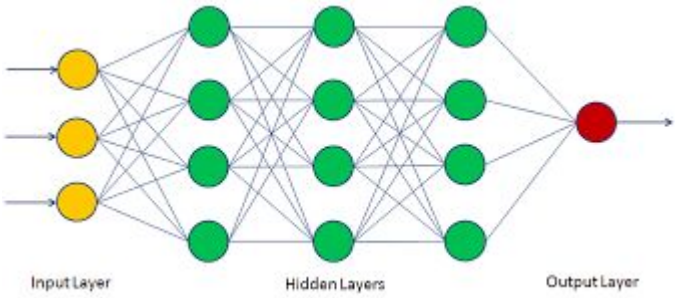
Model

TRANSFORMER

Input = T×224 (T windows × 224 channels)
Learns temporal patterns
Outputs a compact sequence representation.

+

MLP



Input = the Transformer summary vector.
Maps it to 5 continuous outputs (forces per finger).

Training

Data splits:

- Train/Val (within-subject): For each subject except S02, used 80% of windows for training and 20% for validation.
  → Validation checks performance on the same subjects as training but on unseen windows.

- Test on S02 to measure generalization to a new subject.

- Fine-tuning: Fine-tuned on 20% of S02 and re-evaluated on the remaining 80%.

RESULTS:

- **The model explains 98.1% of the variance on the validation set (in Newtons). (Val R^2 = 0.981)**
- **The average RMSE is 2.31% of each finger's dynamic range. (Val NRMSE% = 2.31%)**

  Meaning: If a finger's range is 10 N, that implies RMSE ≈ 0.0231 × 10 ≈ 0.23 N

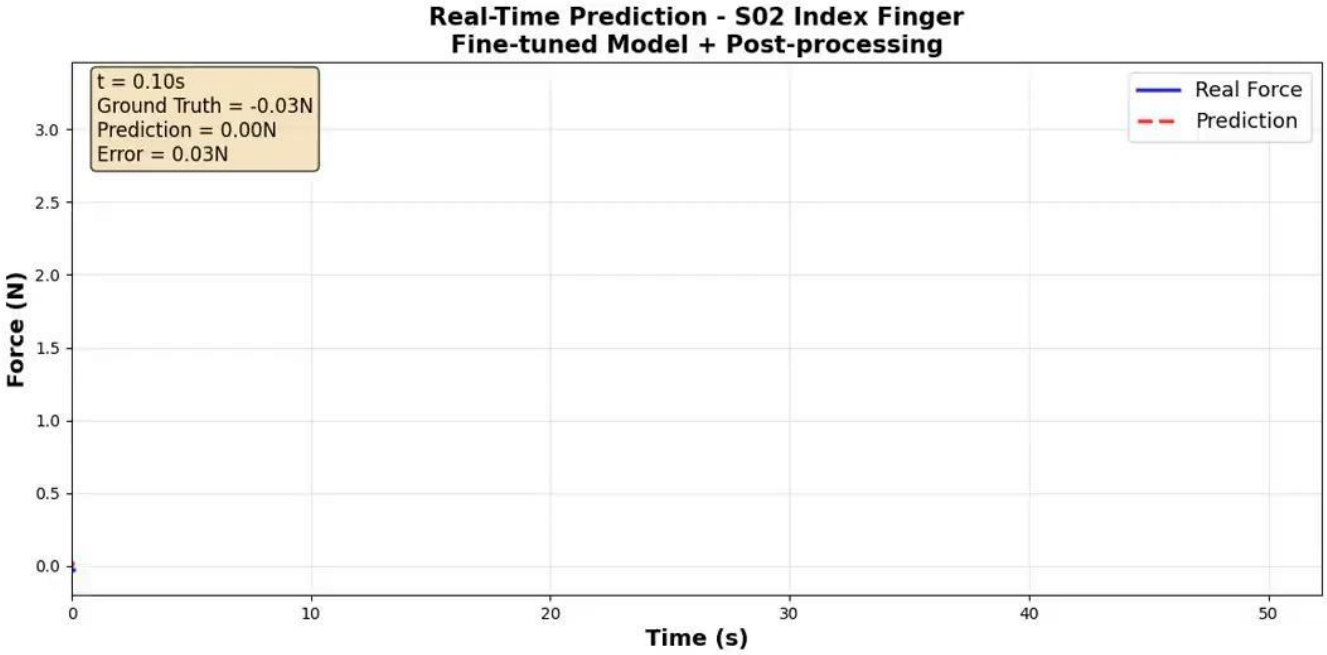
Post Processing

[18]: “The results showed that the post-processing algorithm increased the recognition accuracy by 41.86%”

1. Shifts and stretches the prediction so its zeros are at zero and its peaks match real Newtons.
2. Forces the output to 0 when it’s only tiny noise; only turns “on” when it clearly rises.
3. No “teleporting”
4. Smoothing

[18]: Lorena Isabel Barona López, CNN-LSTM and post-processing for EMG-based hand gesture recognition

DEMO on a subject with fine tuning



Futur work

1. Implement Online predictions
2. Implement the new process for MFT
3. Give precise feedback for clinicians
4. What should we do? What should we write?